

Django Lab 2

Django Template Language (DTL) & Static CSS

Lab Requirements

This project demonstrates Django Template Language using variables, filters, and template tags, together with CSS served through Django static files.

Requirement	Implementation
Study Material	Django Template Language (DTL) concepts
Project 1	Student Portfolio
Project 2	Product Store
Project 3	Number Guessing Game
DTL	Variables, filters, for loops, if/else conditions, empty handling
CSS	External stylesheet loaded with Django {% static %}

Project 1 — Student Portfolio

A simple portfolio page displaying student information and skills dynamically.

DTL examples:

```
{{ student_name|upper }}
```

The **upper** filter converts the student's name to uppercase.

```
{% for skill in skills %} ... {% empty %} ... {% endfor %}
```

The **for** tag loops through the skills list and the **empty** clause handles an empty list.

```
{% if is_available %} ... {% else %} ... {% endif %}
```

The **if/else** tags display availability status.

Project 2 — Product Store

A dynamic product listing containing product name, category, price, discount, and stock status.

DTL examples:

```
{{ product.category|lower }}
```

Uses the **lower** filter to format the category.

```
{{ product.price|floatformat:2 }}
```

Uses **floatformat** to show prices with two decimal places.

```
{% for product in products %} ... {% endfor %}
```

Loops through all products and uses conditional logic to display stock status.

Project 3 — Number Guessing Game

An interactive page where the user enters a number from 1 to 10. The Django view processes the query and the template displays the result.

Template logic:

```
{% if message %} ... {% else %} ... {% endif %}
```

Conditional rendering displays a success message or an instruction.

```
{% for number in numbers %} {{ number }} {% endfor %}
```

A template loop displays the available numbers from 1 to 10.

Static CSS

The project uses an external stylesheet located at **static/css/style.css**.

```
{% load static %}  
<link rel="stylesheet" href="{% static 'css/style.css' %}">
```

The stylesheet provides responsive cards, navigation, buttons, forms, status badges, and mobile-friendly layouts.

Project Structure

```
Django_Lab2/  
  ■■■ manage.py  
  ■■■ requirements.txt  
  ■■■ README.md  
  ■■■ lab2/  
    ■ ■■■ settings.py  
    ■ ■■■ urls.py  
    ■ ■■■ wsgi.py  
  ■■■ main/  
    ■ ■■■ apps.py  
    ■ ■■■ views.py  
    ■ ■■■ migrations/  
  ■■■ templates/  
    ■ ■■■ base.html  
    ■ ■■■ home.html  
    ■ ■■■ portfolio.html  
    ■ ■■■ store.html  
    ■ ■■■ game.html  
  ■■■ static/  
  ■■■ css/  
  ■■■ style.css
```

How to Run

Install dependencies:

```
pip install -r requirements.txt
```

Run migrations:

```
python manage.py migrate
```

Start the development server:

```
python manage.py runserver
```

Then open <http://127.0.0.1:8000/> in a browser.